

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
```

```
df = pd.read_csv("marketing_sales_data.csv")
```

```
df.head()
```

	TV	Radio	Social Media	Influencer	Sales
0	Low	3.518070	2.293790	Micro	55.261284
1	Low	7.756876	2.572287	Mega	67.574904
2	High	20.348988	1.227180	Micro	272.250108
3	Medium	20.108487	2.728374	Mega	195.102176
4	High	31.653200	7.776978	Nano	273.960377

```
print(df.shape)
```

```
df.info()
```

```
df.describe()
```

```
(572, 5)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0    TV              572 non-null    object
1    Radio           572 non-null    float64
2    Social Media    572 non-null    float64
3    Influencer      572 non-null    object
4    Sales           572 non-null    float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB
```

	Radio	Social Media	Sales
count	572.000000	572.000000	572.000000
mean	17.520616	3.333803	189.296908
std	9.290933	2.238378	89.871581
min	0.109106	0.000031	33.509810
25%	10.699556	1.585549	118.718722
50%	17.149517	3.150111	184.005362
75%	24.606396	4.730408	264.500118
max	42.271579	11.403625	357.788195

```
df.isnull().sum()
```

```
TV          0
Radio       0
Social Media 0
Influencer  0
Sales       0
dtype: int64
```

```
print("Total Missing Values:", df.isnull().sum().sum())
```

```
Total Missing Values: 0
```

```
df_encoded = pd.get_dummies(
    df,
    columns=["TV", "Influencer"],
```

```

    drop_first=True
)

df_encoded = df_encoded.astype(float)

df_encoded.head()

```

	Radio	Social Media	Sales	TV_Low	TV_Medium	Influencer_Mega	Influencer_Micro	Influencer_Nano
0	3.518070	2.293790	55.261284	1.0	0.0	0.0	1.0	0.0
1	7.756876	2.572287	67.574904	1.0	0.0	1.0	0.0	0.0
2	20.348988	1.227180	272.250108	0.0	0.0	0.0	1.0	0.0
3	20.108487	2.728374	195.102176	0.0	1.0	1.0	0.0	0.0
4	31.653200	7.776978	273.960377	0.0	0.0	0.0	0.0	1.0

```

corr_matrix = df_encoded.corr()

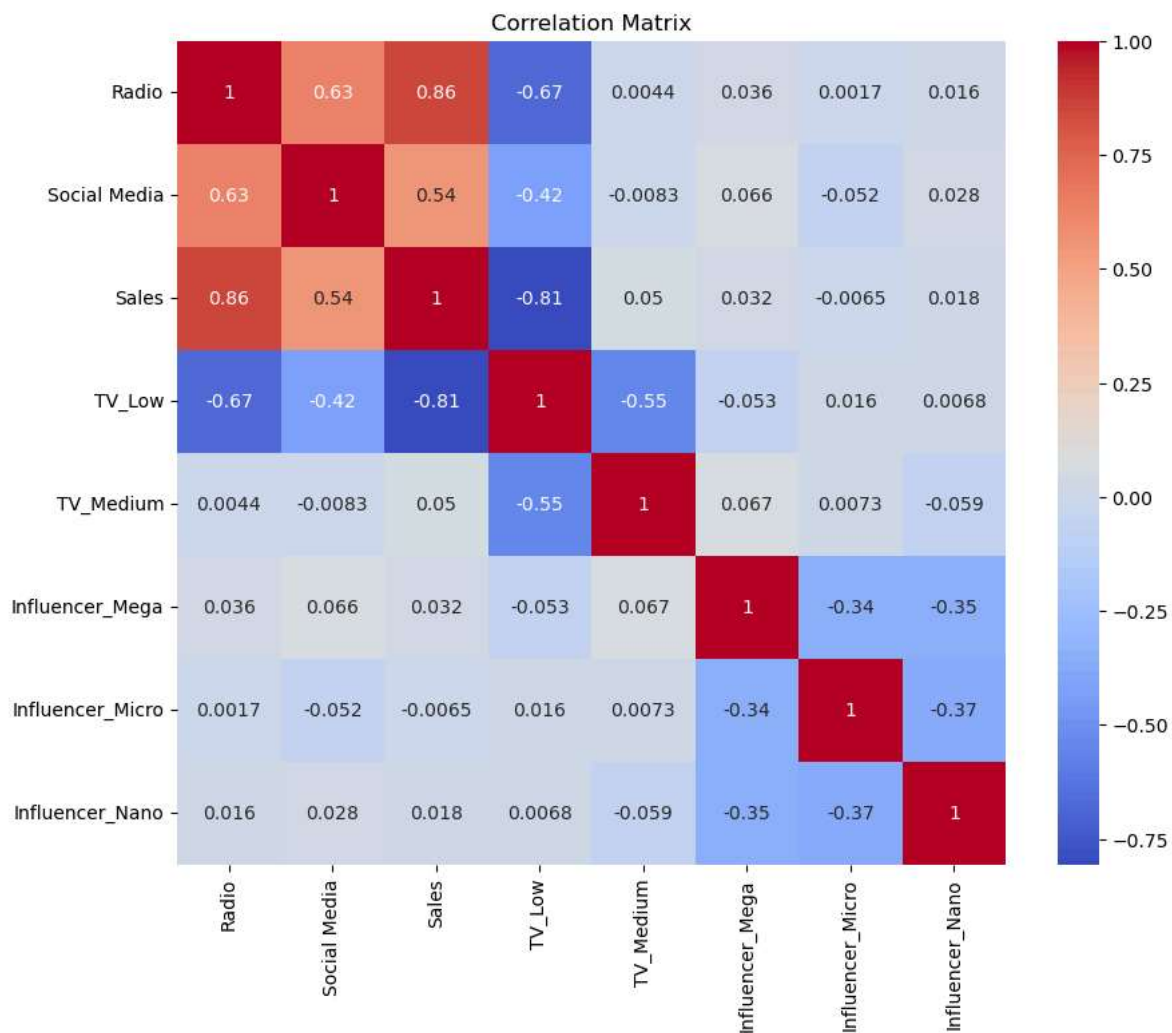
plt.figure(figsize=(10,8))

sns.heatmap(
    corr_matrix,
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

plt.show()

```



```
X_vif = df_encoded.drop("Sales", axis=1)
```

```
X_vif = sm.add_constant(X_vif)

vif = pd.DataFrame()

vif["Feature"] = X_vif.columns

vif["VIF"] = [
    variance_inflation_factor(
        X_vif.values,
        i
    )
    for i in range(X_vif.shape[1])
]

print(vif)
```

	Feature	VIF
0	const	31.649565
1	Radio	3.479641
2	Social Media	1.669098
3	TV_Low	4.076384
4	TV_Medium	2.233350
5	Influencer_Mega	1.593889
6	Influencer_Micro	1.618434
7	Influencer_Nano	1.627430

```
X = df_encoded.drop("Sales", axis=1)

y = df_encoded["Sales"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          Sales      R-squared:                0.904
Model:                  OLS       Adj. R-squared:            0.903
Method:                 Least Squares   F-statistic:           760.4
Date:                  Wed, 17 Jun 2026   Prob (F-statistic):     1.82e-282
Time:                  11:34:20    Log-Likelihood:        -2713.4
No. Observations:      572          AIC:                   5443.
Df Residuals:          564          BIC:                   5478.
Df Model:               7
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                217.4784      6.584      33.031      0.000      204.546      230.411
Radio                 2.9735      0.235      12.644      0.000       2.512       3.435
Social Media         -0.1391      0.676      -0.206      0.837      -1.467       1.189
TV_Low              -154.5736      4.949     -31.231      0.000     -164.295     -144.852
TV_Medium            -75.5947      3.647     -20.726      0.000     -82.759     -68.431
Influencer_Mega       2.4948      3.462      0.721      0.471      -4.305       9.295
Influencer_Micro      2.9391      3.378      0.870      0.385      -3.695       9.574
Influencer_Nano       0.8015      3.346      0.240      0.811      -5.770       7.373
=====
Omnibus:              58.711    Durbin-Watson:           1.874
Prob(Omnibus):        0.000    Jarque-Bera (JB):        17.802
Skew:                 0.057    Prob(JB):                0.000136
Kurtosis:             2.143    Cond. No.                 147.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
print("Adjusted R-Squared:", model.rsquared_adj)
```

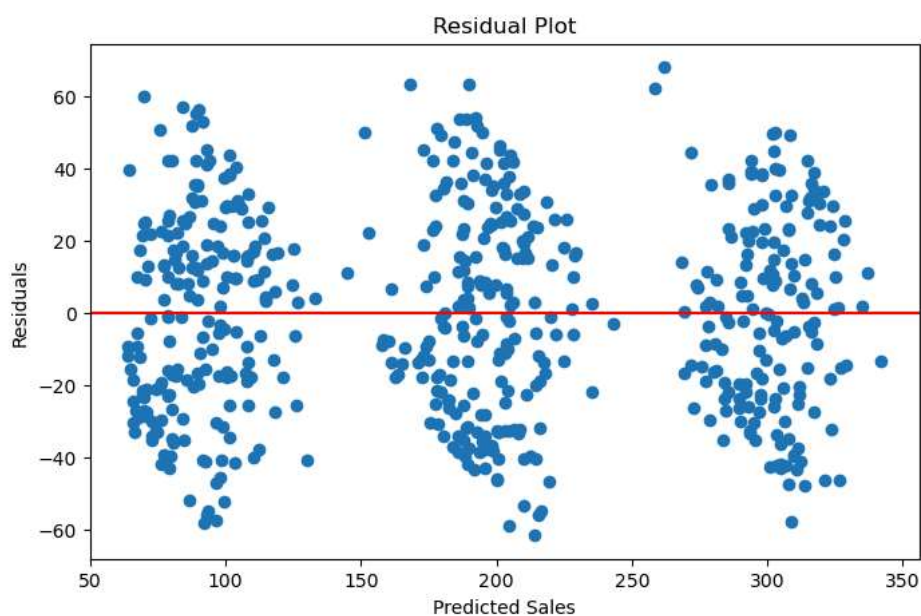
Adjusted R-Squared: 0.9030011006551627

```
predictions = model.predict(X)

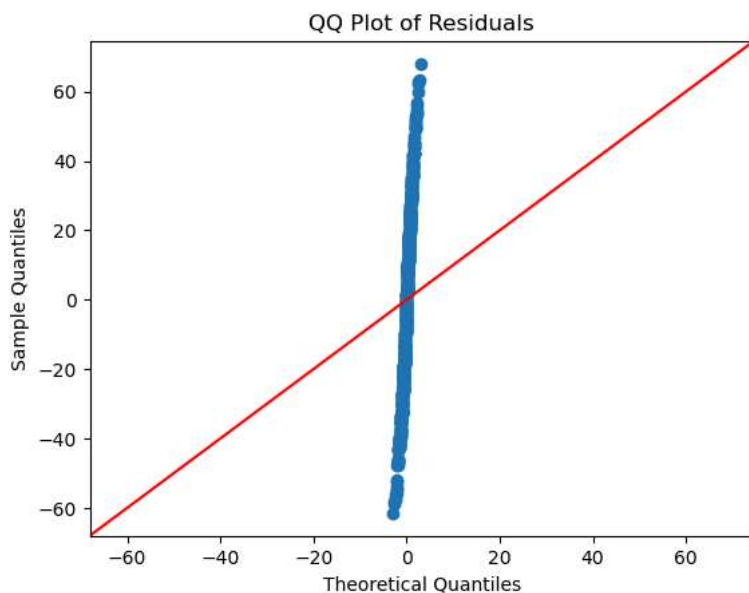
residuals = y - predictions
```

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(  
    predictions,  
    residuals  
)  
  
plt.axhline(  
    y=0,  
    color="red"  
)  
  
plt.xlabel("Predicted Sales")  
plt.ylabel("Residuals")  
  
plt.title("Residual Plot")  
  
plt.show()
```



```
sm.qqplot(  
    residuals,  
    line="45"  
)  
  
plt.title("QQ Plot of Residuals")  
  
plt.show()
```



F-Statistic Interpretation

The model F-statistic is 760.4 with a p-value of 1.82e-282. Since the p-value is significantly less than 0.05, the overall regression model is statistically significant. This indicates that the predictors collectively provide meaningful explanatory power for Sales.

Business Recommendation

The regression model achieved an Adjusted R-squared of 0.903, indicating that approximately 90.3% of the variation in Sales is explained by the included marketing variables.

Radio advertising has a positive and statistically significant effect on Sales. TV advertising also plays a significant role, with the High TV category outperforming Medium and Low TV campaigns.

Social Media and Influencer categories were not statistically significant predictors in this model. Therefore, marketing budget should prioritize TV and Radio advertising channels while reassessing spending on Social Media and Influencer campaigns.